

Deep Learning Curriculum

Beginner → Advanced | Structured learning roadmap

Goal: Build a strong practical foundation in neural networks, PyTorch, CNNs, modern architectures, training, optimization, and deployment.

LEVEL 1 — DEEP LEARNING FUNDAMENTALS

1. Introduction to Deep Learning

- What is Deep Learning?
- Machine Learning vs Deep Learning
- Neural networks
- Artificial neurons
- Biological neuron — basic intuition
- Features and representations
- Forward propagation
- Training vs inference

2. Neural Network Fundamentals

- Perceptron
- Single-layer neural network
- Multi-layer neural network
- Input layer
- Hidden layers
- Output layer
- Weights
- Bias
- Parameters
- Activations

3. Activation Functions

- Why activation functions are needed
- Linear
- Sigmoid
- Tanh
- ReLU
- Leaky ReLU
- ELU
- GELU
- Softmax

LEVEL 2 — NEURAL NETWORK TRAINING

4. Forward Propagation

- Layer computation
- Matrix multiplication
- Activation
- Network output
- Computational graph

5. Loss Functions

- What is a loss function?
- MSE
- MAE
- Binary Cross Entropy
- Categorical Cross Entropy
- Cross Entropy
- Loss calculation

6. Backpropagation

- Why backpropagation
- Computational graphs
- Chain rule
- Gradients
- Gradient calculation
- Weight updates

7. Gradient Descent

- Batch gradient descent
- Stochastic gradient descent
- Mini-batch gradient descent
- Learning rate
- Learning-rate schedules

8. Optimizers

- SGD
- Momentum
- AdaGrad
- RMSProp
- Adam
- AdamW

LEVEL 3 — BUILDING NEURAL NETWORKS

9. Neural Network Architecture

- Fully connected layers
- Hidden layers
- Depth vs width
- Parameters
- Model capacity

10. Training Process

- Epoch
- Batch
- Iteration
- Training loop
- Validation loop
- Checkpoints
- Model saving/loading

11. Weight Initialization

- Zero initialization
- Random initialization
- Xavier / Glorot
- He initialization
- Initialization problems

12. Normalization

- Batch Normalization
- Layer Normalization
- Instance Normalization
- Group Normalization
- When normalization is useful

LEVEL 4 — GENERALIZATION & REGULARIZATION

13. Overfitting

- Underfitting
- Overfitting
- Generalization
- Model capacity

14. Regularization

- L1
- L2 / Weight decay
- Dropout
- Early stopping
- Data augmentation
- Label smoothing

15. Training Diagnostics

- Training loss curves
- Validation loss curves
- Accuracy curves
- Learning-rate behavior
- Detecting overfitting
- Detecting underfitting
- Error analysis

LEVEL 5 — PYTORCH

16. PyTorch Fundamentals

- Installation
- Tensors
- Tensor shapes
- Tensor indexing
- Tensor operations
- Broadcasting
- CPU vs GPU

17. Autograd

- Automatic differentiation
- Computational graphs
- requires_grad
- backward()
- Gradients

18. Neural Network API

- nn.Module
- nn.Linear
- Activation layers
- Sequential models
- Custom models

19. Dataset & DataLoader

- Dataset
- DataLoader
- Batch processing
- Shuffling
- Custom datasets
- Transforms

20. Training in PyTorch

- Training loop
- Validation loop
- Optimizers
- Loss functions
- Backpropagation
- Checkpoints
- GPU training

21. PyTorch Model Management

- state_dict
- Saving/loading models
- Checkpointing
- Evaluation mode
- Inference mode

LEVEL 6 — CONVOLUTIONAL NEURAL NETWORKS

22. CNN Fundamentals

- Why CNNs
- Convolution
- Kernel/filter
- Stride
- Padding
- Feature maps
- Receptive field

23. Pooling

- Max pooling
- Average pooling
- Global average pooling

24. CNN Architecture

- Convolution blocks
- Stacking convolution layers
- Channels
- Spatial dimensions
- Feature hierarchy

25. Important CNN Architectures

- LeNet
- AlexNet
- VGG
- GoogLeNet / Inception
- ResNet
- DenseNet
- MobileNet
- EfficientNet

LEVEL 7 — TRANSFER LEARNING

26. Transfer Learning

- Pretrained models
- Feature extraction
- Fine-tuning
- Freezing layers
- Unfreezing layers
- Learning-rate selection

27. Model Fine-Tuning

- Full fine-tuning
- Partial fine-tuning
- Differential learning rates
- Fine-tuning strategies

LEVEL 8 — DATA & TRAINING TECHNIQUES

28. Data Augmentation

- Why augmentation
- Random crop
- Flip
- Rotation
- Translation
- Scaling
- Color transformations
- Random erasing
- MixUp
- CutMix

29. Dataset Preparation

- Dataset organization
- Train/validation/test datasets
- Annotation formats
- Class distribution
- Data quality

30. *Advanced Training Techniques*

- Learning-rate scheduling
- Warmup
- Gradient clipping
- Gradient accumulation
- Mixed precision
- Automatic Mixed Precision

LEVEL 9 — IMAGE CLASSIFICATION

31. *Classification Pipeline*

- Dataset
- Preprocessing
- Model
- Training
- Validation
- Testing
- Inference

32. *Classification Metrics*

- Accuracy
- Precision
- Recall
- F1-score
- Top-1 accuracy
- Top-5 accuracy
- Confusion matrix

33. *Classification Improvements*

- Class imbalance
- Weighted loss
- Augmentation
- Fine-tuning
- Error analysis

LEVEL 10 — ADVANCED ARCHITECTURES

34. *Residual Networks*

- Residual learning
- Skip connections
- ResNet blocks
- Why residual connections work

35. *Attention*

- Attention concept
- Query
- Key
- Value
- Self-attention
- Multi-head attention

36. Transformers

- Transformer architecture
- Encoder
- Decoder
- Positional encoding
- Self-attention
- Multi-head attention

37. Vision Transformers

- Patch embeddings
- Class token
- Position embeddings
- ViT architecture
- Training ViTs

38. Modern Architectures

- Swin Transformer
- ConvNeXt
- Efficient architectures
- Hybrid CNN/Transformer architectures

LEVEL 11 — GENERATIVE DEEP LEARNING

39. Autoencoders

- Encoder
- Latent representation
- Decoder
- Reconstruction loss
- Denoising autoencoder

40. Variational Autoencoders

- Latent distributions
- Encoder/decoder
- KL divergence
- Reconstruction loss

41. GANs

- Generator
- Discriminator
- Adversarial training
- DCGAN
- Training challenges

42. Diffusion Models

- Forward diffusion
- Noise
- Reverse denoising
- U-Net
- Conditioning
- Diffusion model fundamentals

LEVEL 12 — DEEP LEARNING ENGINEERING

43. GPU Computing

- CUDA basics
- GPU memory
- CPU vs GPU
- Memory transfer
- Batch size
- GPU utilization

44. Performance Optimization

- Mixed precision
- Gradient accumulation
- Efficient DataLoader
- Batch optimization
- Memory optimization

45. Model Optimization

- Quantization
- Pruning
- Knowledge distillation
- Model compression

46. Model Export

- TorchScript
- ONNX
- Model serialization
- Inference optimization

LEVEL 13 — EXPERIMENTATION & PRODUCTION

47. Experiment Tracking

- Experiment organization
- Parameters
- Metrics
- Checkpoints
- MLflow / Weights & Biases

48. Reproducibility

- Random seeds
- Dataset versions
- Configuration management
- Environment management

49. Model Evaluation & Error Analysis

- Evaluation datasets
- Failure cases
- Confidence analysis
- Misclassification analysis
- Model comparison

50. Deployment Fundamentals

- Batch inference
- Real-time inference
- Model serving
- API inference
- Docker
- Monitoring